

# Packet tracer vanaf 0

17 March 2026 14:03

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name SecretariaatKleuters
Switch(config-vlan)#vlan 11
Switch(config-vlan)# name SecretariaatMiddelbaar
Switch(config-vlan)#vlan 12
Switch(config-vlan)# name SecretariaatLager
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name KlassenIsteVerdiepMiddelbaar
Switch(config-vlan)#vlan 21
Switch(config-vlan)# name KlassenGelijkvloers
Switch(config-vlan)#vlan 22
Switch(config-vlan)# name KlassenLager
Switch(config-vlan)#vlan 23
Switch(config-vlan)# name KlassenKleuter
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name Servers
Switch(config-vlan)#vlan 40
Switch(config-vlan)# name WifiGelijkvloers
Switch(config-vlan)#vlan 41
Switch(config-vlan)# name WifiGelijkvloers2
Switch(config-vlan)#vlan 42
Switch(config-vlan)# name WifiIsteVerdiep
Switch(config-vlan)#vlan 43
Switch(config-vlan)# name WifiIsteVerdiep2
Switch(config-vlan)#vlan 44
Switch(config-vlan)# name WifiDocentenMiddel
Switch(config-vlan)#vlan 99
Switch(config-vlan)# name Management
Switch(config-vlan)#end
Switch#write memory
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%1:INFORMED-6-1:DOWN- line protocol on Interface GigabitEthernet0/1.
```

## Alle Vlan's opnieuw aangemaakt op alle switches

SYS-5-CONFIG\_I: Configured from console by console

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int vlan 99
Switch(config-if)#ip add 10.2.99.12 255.255.255.0
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
LINK-5-CHANGED: Interface Vlan99, changed state to up
```

SYS-5-CONFIG\_I: Configured from console by console

## IP's op VLAN99 gezet (management)

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SKleuterKlas
SKleuterKlas(config)#
```

## Hostname's veranderen op alle switches

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 10.2.10.1 10.2.10.5
Router(config)#ip dhcp excluded-address 10.2.10.33 10.2.10.35
Router(config)#ip dhcp excluded-address 10.2.10.65 10.2.10.67
Router(config)#ip dhcp excluded-address 10.2.20.1 10.2.20.10
Router(config)#ip dhcp excluded-address 10.2.21.129 10.2.21.132
Router(config)#ip dhcp excluded-address 10.2.21.193 10.2.21.198
Router(config)#ip dhcp excluded-address 10.2.22.1 10.2.22.5
Router(config)#ip dhcp excluded-address 10.2.30.1 10.2.30.10
Router(config)#end
Router#wr
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Statische adressen in de router toewijzen zodat deze als gateway/demo pc kunnen dienen, dit deed ik ook bij de MLS

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#ip dhcp pool VLAN10
Switch(dhcp-config)# network 10.2.10.0 255.255.255.224
Switch(dhcp-config)# default-router 10.2.10.1
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN11
Switch(dhcp-config)# network 10.2.10.32 255.255.255.224
Switch(dhcp-config)# default-router 10.2.10.33
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN12
Switch(dhcp-config)# network 10.2.10.64 255.255.255.224
Switch(dhcp-config)# default-router 10.2.10.65
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN20
Switch(dhcp-config)# network 10.2.20.0 255.255.255.0
Switch(dhcp-config)# default-router 10.2.20.1
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN21
Switch(dhcp-config)# network 10.2.21.192 255.255.255.224
Switch(dhcp-config)# default-router 10.2.21.193
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN22
Switch(dhcp-config)# network 10.2.22.0 255.255.255.0
Switch(dhcp-config)# default-router 10.2.22.1
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool VLAN23
Switch(dhcp-config)# network 10.2.21.128 255.255.255.192
Switch(dhcp-config)# default-router 10.2.21.129
Switch(dhcp-config)# dns-server 8.8.8.8
Switch(dhcp-config)#exit
```

#### DHCP pools aangemaakt op de MLS alsook het instellen van de default-gateway

```
Switch(config)#
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range gig0/1 - 2
Switch(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on
the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on
the remote port.
Switch(config-if-range)#exit
Switch(config)#interface Port-channel 1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan all
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#no shutdown end
Creating a port-channel interface Port-channel 1
```

#### Etherchannel's aanmaken vanaf de distribution switches naar MLS en van acces-layer switch in het lager naar de distribution switches

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no in
Switch(config)#no interface po
Switch(config)#no interface port-channel 1
Switch(config)#no in
Switch(config)#no interface po
Switch(config)#no interface port-channel 2
Switch(config)#
```

Port channel Op de access-layer switches 1 access poort gezet voor de desbetreffend lokaal voor vaste pc's vlan 40, Windows en wetenschap vlan 50, Linux = vlan 60, administratie = vlan 25 -> het desbetreffende lokaal worden.

```
Switch(config)#int range fa0/3 - 24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 21
Switch(config-if-range)#no shutdown
```

#### VLAN access toegewezen op de access-layer

```

MLS>en
MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLS(config)#vlan 10
MLS(config-vlan)# name Studenten
MLS(config-vlan)#vlan 20
MLS(config-vlan)# name Docenten
MLS(config-vlan)#vlan 25
MLS(config-vlan)# name Administratie
MLS(config-vlan)#vlan 30
MLS(config-vlan)# name Guest
MLS(config-vlan)#vlan 40
MLS(config-vlan)# name Klaslokalen
MLS(config-vlan)#vlan 50
MLS(config-vlan)# name Wetenschap
MLS(config-vlan)#vlan 60
MLS(config-vlan)# name Linux
MLS(config-vlan)#vlan 70
MLS(config-vlan)# name IT-Admin
MLS(config-vlan)#vlan 80
MLS(config-vlan)# name Servers
MLS(config-vlan)#vlan 99
MLS(config-vlan)# name Management
MLS(config-vlan)#

```

Nieuwe VLAN's ingesteld om dit praktischer te maken voor ACL's

```

Klassen1stel(config)#spanning-tree vlan 1-4094 priority 40960
Klassen1stel(config)#

```

Alle access-layer switches in het middelbaar spanning-tree aware gemaakt, met normale priority

```

MLS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLS(config)#spanning-tree vlan 1-4094 root primary
MLS(config)#

```

MLS root primary gemaakt voor stp

```

SWDistrMid1(config)#spanning-tree vlan 1-4094 root secondary
SWDistrMid1(config)#end
SWDistrMid1#
%SYS-5-CONFIG_I: Configured from console by console

```

Eerste distributie switch van het middelbaar de root secondary gemaakt, ter voorbereiding op een etherchannel tussen distributie 1 en 2

```

SWDistrMid2(config)#interface range fa0/21 - 22
SWDistrMid2(config-if-range)#switchport mode trunk
SWDistrMid2(config-if-range)#switchport trunk native vlan 999
SWDistrMid2(config-if-range)#switchport trunk allowed vlan
10,20,25,30,40,50,60,70,80,99,999
SWDistrMid2(config-if-range)#channel-group 5 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
SWDistrMid2(config-if-range)#no shutdown
SWDistrMid2(config-if-range)#exit
SWDistrMid2(config)#interface port-channel 5
SWDistrMid2(config-if)#switchport mode trunk
SWDistrMid2(config-if)#switchport trunk native vlan 999
SWDistrMid2(config-if)#switchport trunk allowed vlan 10,20,25,30,40,50,60,70,80,99,999
SWDistrMid2(config-if)#exit
Creating a port-channel interface Port-channel 5

```

Een etherchannel opgesteld tussen Distribution switch 1 en 2

```

SWDistrMid1(config)#interface range fa0/12-16
SWDistrMid1(config-if-range)#swi
SWDistrMid1(config-if-range)#switchport mo
SWDistrMid1(config-if-range)#switchport mode ac
SWDistrMid1(config-if-range)#switchport mode access
SWDistrMid1(config-if-range)#sw
SWDistrMid1(config-if-range)#switchport ac
SWDistrMid1(config-if-range)#switchport access v
SWDistrMid1(config-if-range)#switchport access vlan 80
SWDistrMid1(config-if-range)#no shutdown
SWDistrMid1(config-if-range)#exit
SWDistrMid1(config)#int fa0/11
SWDistrMid1(config-if)#int
SWDistrMid1(config-if)#sw
SWDistrMid1(config-if)#switchport mo
SWDistrMid1(config-if)#switchport mode ac
SWDistrMid1(config-if)#switchport mode access
SWDistrMid1(config-if)#sw
SWDistrMid1(config-if)#switchport a
SWDistrMid1(config-if)#switchport access v
SWDistrMid1(config-if)#switchport access vlan 70
SWDistrMid1(config-if)#no shutdown
SWDistrMid1(config-if)#exit
SWDistrMid1(config)#

```

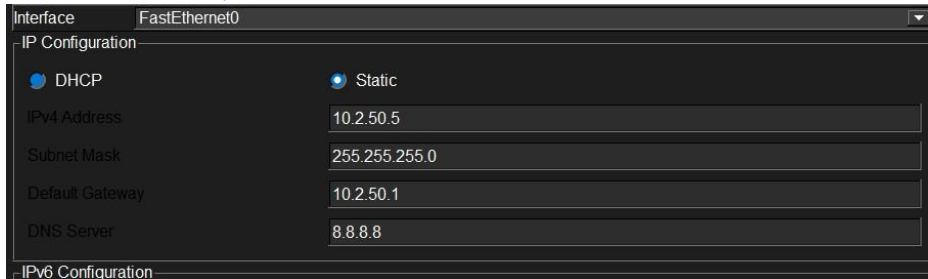
Access ports toegevoegd voor de servers op distributie switch 1 (VLAN80) alsook 1 access port op VLAN 70 voor it-admin om ACL's te kunnen tonen later

```

Klassen1stel(config)#int fa0/1
Klassen1stel(config-if)#swi
Klassen1stel(config-if)#switchport mo
Klassen1stel(config-if)#switchport mode ac
Klassen1stel(config-if)#switchport mode access
Klassen1stel(config-if)#swi
Klassen1stel(config-if)#switchport ac
Klassen1stel(config-if)#switchport access v
Klassen1stel(config-if)#switchport access vlan 40
Klassen1stel(config-if)#no shutdown
Klassen1stel(config-if)#exit

```

Op de access-layer switches 1 access poort gezet voor de desbetreffend lokaal voor vaste pc's vlan 40, Windows en wetenschap vlan 50, Linux = vlan 60, administratie = vlan 25



Nadien statische IP's toegewezen aan des betreffende PC's die overeen komen met hun VLAN.

```

SWDistrMidl(config)#int fa0/17
SWDistrMidl(config-if)#swi
SWDistrMidl(config-if)#switchport mo
SWDistrMidl(config-if)#switchport mode ac
SWDistrMidl(config-if)#switchport mode access
SWDistrMidl(config-if)#swi
SWDistrMidl(config-if)#switchport ac
SWDistrMidl(config-if)#switchport access v
SWDistrMidl(config-if)#switchport access vlan 10
SWDistrMidl(config-if)#sp
SWDistrMidl(config-if)#span
SWDistrMidl(config-if)#spanning-tree po
SWDistrMidl(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

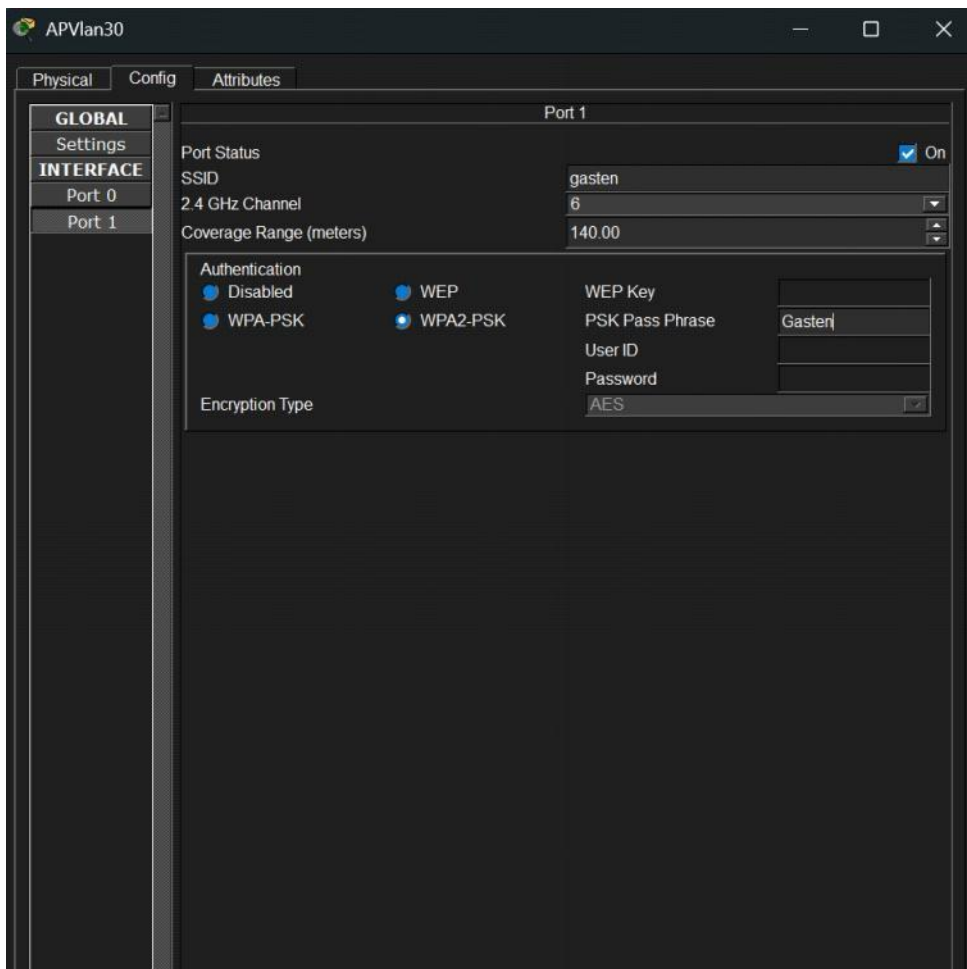
%Portfast has been configured on FastEthernet0/17 but will only
have effect when the interface is in a non-trunking mode.
SWDistrMidl(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to down
SWDistrMidl(config-if)#no shu
SWDistrMidl(config-if)#no shutdown
SWDistrMidl(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to up

```

AP's toegang gegeven tot hun specifieke VLAN; zo blijven ACL's testen later makkelijk. Dit betekent dus 3 AP's in packet tracer (VLAN10,20 en 30) per echte AP dit heb ik 2 keer ingesteld





Een logische SSID aan de wifi AP's gegeven en een passphrase om pingen te kunnen testen en later te bekijken of ACL's werken.

```
KlassenGelijkvl(config)#ip domain-name Dennenappel.local
KlassenGelijkvl(config)#crypto key generate rsa
The name for the keys will be: KlassenGelijkvl.Dennenappel.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

KlassenGelijkvl(config)#username admin privilege 15 secret Studentpxl
*Mar 1 5:45:26.804: %SSH-5-ENABLED: SSH 2 has been enabled
KlassenGelijkvl(config)#ip ssh version 2
KlassenGelijkvl(config)#line vty 0 15
KlassenGelijkvl(config-line)# transport input ssh
KlassenGelijkvl(config-line)# login local
KlassenGelijkvl(config-line)# exit
KlassenGelijkvl(config)#end
KlassenGelijkvl#wr
%SYS-5-CONFIG_I: Configured from console by console
```

SSH geconfigureerd op iedere switch zodat deze bereikbaar zijn via VLAN70 wat de IT-Admin's zijn

Port security toegevoegd op de Distribution switches. Dit voorkomt het aansluiten van extra switches en zet alle ongebruikte poorten automatisch in err-disabled. Dhcp snooping zorgt dat er geen ongewenste dhcp servers op het netwerk komen. Spanning-tree guard root voorkomt ook dat andere trunks de root bridge zouden worden.

```
SWKlassenkleuter(config)#int range fa0/1 - 23
SWKlassenkleuter(config-if-range)#shut
```

Manueel alle ongebruikte poorten uitgezet op alle access-layer switches. Hetzelfde deed ik voor de MLS

Op alle access-ports gezorgd dat er een gelimiteerd aantal MAC-adressen geleerd kunnen worden, voorkomt dynamisch trunksen, en skipt STP listening. Alle AP's juist ingesteld met SSID en

```
SWDistrMid2(config)#interface range Gi0/1 - 2
SWDistrMid2(config-if-range)# ip dhcp snooping trust
SWDistrMid2(config-if-range)# ip arp inspection trust
SWDistrMid2(config-if-range)# switchport trunk native vlan 999
SWDistrMid2(config-if-range)# switchport trunk allowed vlan 10,20,25,30,40,50,60,70,80,99
SWDistrMid2(config-if-range)# switchport nonegotiate
SWDistrMid2(config-if-range)# spanning-tree guard root
SWDistrMid2(config-if-range)#end
```

passphrases

```
KlassenGelijkvl(config)#interface range Fa0/1
KlassenGelijkvl(config-if-range)#switchport mode access
KlassenGelijkvl(config-if-range)#switchport port-security
KlassenGelijkvl(config-if-range)#switchport port-security maximum 2
KlassenGelijkvl(config-if-range)#switchport port-security mac-address sticky
KlassenGelijkvl(config-if-range)#switchport port-security violation shutdown
KlassenGelijkvl(config-if-range)#switchport nonegotiate
KlassenGelijkvl(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
KlassenGelijkvl(config-if-range)#switchport access vlan 40
KlassenGelijkvl(config-if-range)#exit
KlassenGelijkvl(config)#
```

```

Router>en
Router#show a
Router#show acc
Router#show access-lists
Extended IP access list VLAN10-IN
 10 deny ip 10.2.8.0 0.0.7.255 10.2.20.0 0.0.0.255
 20 deny ip 10.2.8.0 0.0.7.255 10.2.25.0 0.0.0.255
 30 deny ip 10.2.8.0 0.0.7.255 10.2.30.0 0.0.0.255
 40 deny ip 10.2.8.0 0.0.7.255 10.2.70.0 0.0.0.255
 50 deny ip 10.2.8.0 0.0.7.255 10.2.99.0 0.0.0.255
 60 permit ip any any
Extended IP access list VLAN20-IN
 10 deny ip 10.2.20.0 0.0.0.255 10.2.30.0 0.0.0.255
 20 deny ip 10.2.20.0 0.0.0.255 10.2.70.0 0.0.0.255
 30 deny ip 10.2.20.0 0.0.0.255 10.2.99.0 0.0.0.255
 40 permit ip any any
Extended IP access list VLAN25-IN
 10 deny ip 10.2.25.0 0.0.0.255 10.2.30.0 0.0.0.255
 20 deny ip 10.2.25.0 0.0.0.255 10.2.70.0 0.0.0.255
 30 deny ip 10.2.25.0 0.0.0.255 10.2.99.0 0.0.0.255
 40 permit ip any any
Extended IP access list VLAN30-IN
 10 deny ip 10.2.30.0 0.0.0.255 10.0.0.0 0.255.255.255
 20 permit ip any any
Extended IP access list VLAN40-IN
 10 deny ip 10.2.40.0 0.0.0.255 10.2.20.0 0.0.0.255
 20 deny ip 10.2.40.0 0.0.0.255 10.2.25.0 0.0.0.255
 30 deny ip 10.2.40.0 0.0.0.255 10.2.30.0 0.0.0.255
 40 deny ip 10.2.40.0 0.0.0.255 10.2.70.0 0.0.0.255
 50 deny ip 10.2.40.0 0.0.0.255 10.2.99.0 0.0.0.255
 60 permit ip any any
Extended IP access list VLAN50-IN
 10 deny ip 10.2.50.0 0.0.0.255 10.2.20.0 0.0.0.255
 20 deny ip 10.2.50.0 0.0.0.255 10.2.25.0 0.0.0.255
 30 deny ip 10.2.50.0 0.0.0.255 10.2.30.0 0.0.0.255
 40 deny ip 10.2.50.0 0.0.0.255 10.2.70.0 0.0.0.255
 50 deny ip 10.2.50.0 0.0.0.255 10.2.99.0 0.0.0.255
 60 permit ip any any
Extended IP access list VLAN60-IN
 10 deny ip 10.2.60.0 0.0.0.255 10.2.20.0 0.0.0.255
 20 deny ip 10.2.60.0 0.0.0.255 10.2.25.0 0.0.0.255
 30 deny ip 10.2.60.0 0.0.0.255 10.2.30.0 0.0.0.255
 40 deny ip 10.2.60.0 0.0.0.255 10.2.70.0 0.0.0.255
 50 deny ip 10.2.60.0 0.0.0.255 10.2.99.0 0.0.0.255
 60 permit ip any any
Extended IP access list VLAN80-IN
 10 deny ip 10.2.80.0 0.0.0.255 10.2.8.0 0.0.7.255
 20 deny ip 10.2.80.0 0.0.0.255 10.2.30.0 0.0.0.255
 30 permit ip any any
Extended IP access list VLAN99-IN
 10 deny ip 10.2.99.0 0.0.0.255 10.2.8.0 0.0.7.255
 20 deny ip 10.2.99.0 0.0.0.255 10.2.20.0 0.0.0.255
 30 deny ip 10.2.99.0 0.0.0.255 10.2.25.0 0.0.0.255
 40 deny ip 10.2.99.0 0.0.0.255 10.2.30.0 0.0.0.255
 50 deny ip 10.2.99.0 0.0.0.255 10.2.40.0 0.0.0.255
 60 deny ip 10.2.99.0 0.0.0.255 10.2.50.0 0.0.0.255
 70 deny ip 10.2.99.0 0.0.0.255 10.2.60.0 0.0.0.255
 80 deny ip 10.2.99.0 0.0.0.255 10.2.80.0 0.0.0.255
 90 permit ip any any
Router#

```

Alle functies in verband met inter-vlan routing, dhcp, security en ACL op de router geconfigureerd en de MLS vervangen door een gewone layer 2 switch